Information Architecture

Project Name

CarbonWise

Version: 1.0

Status: Approved

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1. Purpose

This document defines how information is organized, structured, navigated, and presented throughout CarbonWise.

The goal is to ensure users can easily:

- Understand their carbon footprint
- Track sustainability progress
- Reduce emissions through personalized actions

while maintaining:

- Accessibility
- Simplicity
- Discoverability
- Scalability

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2. Information Architecture Goals

Primary Goals

Understand

Help users quickly identify:

- Carbon Score
- Emission Sources
- Sustainability Status

Track

Help users monitor:

- Trends
- Goals
- Historical Data

Reduce

Help users discover:

- Personalized Recommendations
- High Impact Actions
- Future Opportunities

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3. Core Navigation Structure

```
Home
│
├── Dashboard
│
├── Carbon Calculator
│
├── Recommendations
│
├── Goals
│
├── Carbon Twin
│
├── Activity History
│
├── Profile
│
└── Settings
```

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4. User Journey Architecture

First-Time User

Goal

Understand footprint and receive first recommendation.

Flow

```
Landing Page

↓

Register

↓

Onboarding

↓

Initial Carbon Assessment

↓

Carbon Score

↓

Recommendations

↓

Goal Creation

↓

Dashboard
```

Outcome

User understands:

- Current footprint

- Major emission source
- First action to take

Returning User

Goal

Track progress and improve sustainability.

Flow

```
Login

↓

Dashboard

↓

Review Trends

↓

View Recommendations

↓

Log Activities

↓

Track Goals

↓

Review Carbon Twin
```

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5. Content Hierarchy

Information is prioritized according to challenge requirements.

Level 1 (Highest Priority)

Directly solves problem statement.

Carbon Score

Recommendations

Emission Breakdown

Progress Tracking

Level 2

Supports behavior change.

Goals

Challenges

Activity History

Level 3

Supporting information.

Profile

Settings

Help Center

---

6. Dashboard Architecture

The dashboard is the most important page.

Dashboard Layout

```
Dashboard

├── Carbon Score
│
├── Sustainability Rating
│
├── Emission Breakdown
│
├── Top Recommendation
│
├── Goal Progress
│
├── Monthly Trends
│
└── Carbon Twin Preview
```

Carbon Score Card

Displays:

- Current Score
- Monthly Change
- Sustainability Rating

Example

```
Carbon Score: 68

Rating: Improving

Monthly Change:

+8%
```

Emission Breakdown

Displays:

```
Transportation

Food

Energy

Shopping
```

Visualization

Charts:

- Pie Chart
- Bar Chart

Purpose:

Identify hotspots quickly.

Top Recommendation Widget

Displays:

```
Highest Impact Recommendation
```

Includes:

- Estimated Savings
- Difficulty Level
- Quick Action Button

Goal Progress Widget

Displays:

- Goal Completion %
- Remaining Reduction
- Next Milestone

Carbon Twin Widget

Displays:

```
What If Preview
```

Example:

```
Use Public Transport

↓

Potential Reduction

120 kg CO₂/year
```

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7. Carbon Calculator Architecture

Purpose:

Help users understand emissions.

Calculator Structure

```
Carbon Calculator

├── Transportation
│
├── Food
```

```
│
├── Energy
│
└── Shopping
```

Transportation Section

Collects:

- Vehicle Type
- Distance
- Fuel Type
- Frequency

Food Section

Collects:

- Diet Type
- Weekly Consumption
- Meal Patterns

Energy Section

Collects:

- Electricity Usage
- Appliance Usage
- Heating Usage

Shopping Section

Collects:

- Product Type
- Purchase Frequency
- Spending Category

Output

After submission:

```
Carbon Score

↓

Category Breakdown

↓

Recommendations
```

8. Recommendations Architecture

Purpose:

Help users reduce emissions.

Structure

```
Recommendations

├── High Impact
│
├── Medium Impact
│
└── Low Impact
```

Recommendation Card

Contains:

Title

Example:

```
Replace Two Car Trips Weekly
```

Estimated Savings

Example:

```
120 kg CO₂/year
```

Difficulty

Options:

```
Easy

Medium

Hard
```

Action Button

Example:

```
Start Goal
```

Recommendation Prioritization

Order:

```
Impact

↓
```

```
Feasibility

↓

User Preferences

↓

Goal Alignment
```

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9. Goal Architecture

Purpose:

Enable measurable progress.

Structure

```
Goals

├── Active Goals
│
├── Completed Goals
│
└── Milestones
```

Goal Card

Displays:

- Goal Name
- Progress %
- Deadline
- Estimated Savings

Example

```
Reduce Transportation Emissions

Progress:

40%

Target:

20% Reduction
```

---

10. Carbon Twin Architecture

Purpose:

Provide predictive insights.

Structure

```
Carbon Twin

├── Current Lifestyle
│
├── Scenario Builder
│
└── Forecast Results
```

Scenario Builder

Users select:

- Transportation Changes
- Food Changes
- Energy Changes

Forecast Results

Displays:

```
Current Emissions

↓

Projected Emissions

↓

Potential Reduction
```

Example

```
Current:

250 kg CO₂

Future:

180 kg CO₂

Savings:

70 kg CO₂
```

---

11. Activity History Architecture

Purpose:

Support tracking.

Structure

```
Activity History

├── Transportation
│
├── Food
│
├── Energy
│
└── Shopping
```

Features

- Filtering
- Search
- Date Range Selection

---

12. Profile Architecture

Purpose:

Enable personalization.

Structure

```
Profile

├── Personal Information
│
├── Sustainability Preferences
│
├── Goals
│
└── Recommendation Preferences
```

Stored Preferences

Examples:

```
Diet Type

Preferred Transportation

Target Reduction

Notification Settings
```

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13. Mobile Information Architecture

Because most users are mobile-first.

Bottom Navigation

```
Dashboard

Calculator

Recommendations

Goals

Profile
```

Mobile Priority Order

1. Carbon Score
2. Recommendations
3. Goals
4. Tracking

---

14. Accessibility Architecture

Supports judging criteria.

WCAG 2.2 AA

Required.

Semantic HTML

Examples:

```
<header>
<nav>
<main>
<section>
<footer>
```

Keyboard Navigation

Supported.

Screen Readers

Supported.

ARIA Labels

Applied to:

- Buttons
- Forms
- Charts

Color Accessibility

Minimum contrast:

```
4.5:1
```

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15. Search Architecture

Users can search:

- Recommendations
- Goals
- Activities
- Sustainability Topics

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16. Personalization Architecture

The interface adapts based on:

- Carbon Score
- Emission Hotspots
- Goal Status
- Recommendation Adoption

Example

High Transportation User:

Shows:

```
Transportation Recommendations First
```

High Food Impact User:

Shows:

```
Food Recommendations First
```

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17. Future Expansion

Architecture supports:

- Community Challenges
- AI Carbon Coach
- Smart Device Integration
- Sustainability Marketplace

without major restructuring.

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18. Information Architecture Success Criteria

The information architecture is successful when:

✅ Users understand their footprint within 60 seconds

✅ Users find recommendations within 2 clicks

✅ Users can create goals within 1 minute

✅ Users can access Carbon Twin simulations easily

✅ Accessibility standards are met

✅ Navigation remains intuitive

✅ The platform directly supports:

- Understand
- Track
- Reduce

objectives from the challenge statement.